

VLSI/IC Design Reference Handbook

Structured Engineering Notes • Custom IC Flow

1. EDA Vendors & Tool Environment

- **Vendors:** Cadence, Synopsys, Mentor Graphics
- **Focus:** Analog Mixed Signal — Code to GDS [*UNCLEAR: Originally written as "Code to Cab/Cat" - VERIFY context*].
- **Tool Suite:** Full Custom IC Design

Terminal Commands

Command / File	Description / Function
<code>virtuoso</code>	Launches Virtuoso, but leaves the terminal locked.
<code>virtuoso &</code>	Launches Virtuoso and keeps the terminal open to accept other commands.
<code>csh</code>	Language required (C shell / Script) to use the tool.
<code>source <Script file></code>	Executes the script file in the current shell.
<code>.cshrc</code>	Contains all tools paths + license fetching configurations.

2. Full Custom IC Design Flow

"Virtuoso - The Art of Engineering Amplified."

Step	Stage	Primary Tool / Suite	Function
1	Schematic	Virtuoso Schematic Editor	Editors for Schematic, Symbol, etc.
2	Simulation	Multi-Mode Sim / Spectre	Analog circuit simulation

3	Layout	Layout Suite VLE/VXL	Physical layout drafting
4	Physical Verif.	Assura (DRC / LVS)	Checks design rules and connectivity matches
5	RC Extraction	Assura Quantus	Extracts parasitics for post-layout analysis
6	GDSII (Tapeout)	Virtuoso Stream Export	Final layout database file

3. Schematic & Symbol Generation

- **CIW:** Command Interpreter Window.
- **Library Naming Convention:** Must start with a Capital letter.
- **Hierarchy Structure:** Library → Cell → View.

Device Sizing Tip

Use **Multiplier** / **Fingers** to reduce area.

Additional technical note: Splitting a large transistor into multiple fingers reduces gate resistance and parasitic source/drain capacitance while optimizing the aspect ratio for layout constraints.

Symbol Creation Procedure

1. Go to: `Create → cellview → from cellview → OK`
2. In Symbol Generation Options: `Edit accordingly → OK`

4. Layout Design (Layout XL)

- **Launch process:** Launch Layout XL.
- **Connectivity generation:** `Connectivity → Generate → All from source → OK`.

Key Shortcuts & Commands

- `Shift + F` : Display hierarchy.

Additional technical note: This shortcut expands layout hierarchy, showing internal layout contents of instantiated pcells/macros instead of just their bounding boxes.

- `p` : Create Wiring (Path/Wire).
- `Align` : Click on open space → Press `A` → Point.

Pin Placement & Body Ties

- **Pin Placement:** Go to: `Place → Pin Placement`.
- **Integrated Body Tie:** Press `Q` (Parameters) → Parameter → Left Body Tie.
- **Detached Body Tie:** Types available are Left, Right, Top, Bottom.

5. Physical Verification (DRC & LVS)

Design Rule Check (DRC)

- **Technology Setup:**
 1. `Assura → Technology → Install folder in directories column.`
 2. Path/Selection: `Apply ← assura_tech ← Sel tech ← Analog ← Foundry (e.g., 45-nm, 180nm)`
- **Running DRC:**
 1. `Assura → Run DRC → Run Name (Any) → OK`
 2. Technology: Change to `gpdk`.
- **Shortcut:** Rectangle is `R`.

Layout Versus Schematic (LVS)

- `Assura → Run LVS → Run Name`. Select Technology.
- **Debug Tip:** "Google LVS Mismatches" to resolve errors.

Area Calculation

- Path: `Tools → Area & Density Calculator`.

6. RC Extraction (Quantus)

1. Go to: `Assura → Run Quantus`.
2. **Setup tab:**
 - RuleSet: `Typical` (Default for 180nm, 90nm).
 - Output: `Extracted View (Spice (Tabular, Num))`.
 - Library Prefix: `."` (Dot).
3. **Extraction tab:** Extraction Type: `Ref Node (vss)`.
4. **Netlisting tab:** Parasitic cap models → Include as comment.

PDK-Specific Note: The "Parasitic cap models -> Include as comment" setting is marked explicitly as "(only 45)", indicating it is specific to the 45nm GPDK.

7. Post-Layout Simulation

1. **Library Manager (Lib Mang):** Select `tb_file` (Testbench) → `New` → `New cell view` → `Type: config`.
2. **New config:** Use template → `spectre` (spectre simulator template).
3. **View:** Set `myview` to Schematic.
4. **Tree View folder:** Right Click → Set instance → `av_extracted` → `Save`.
5. **Table View:** Resolve any Red lines → `Save`.
6. **Open:** The testbench will now have the `av_extracted` view loaded for simulation.

8. Tapeout / GDSII Export

1. From **CIW:** `File` → `Export` → `Stream`.
2. **Library browser:** Select the layout (e.g., `Nand expl layout`).
3. **Tech lib:** Select `gpdk 45`.
4. **Translate:** Select `yes`.
5. **View:** View all application → `Text Editor` → `Select`.